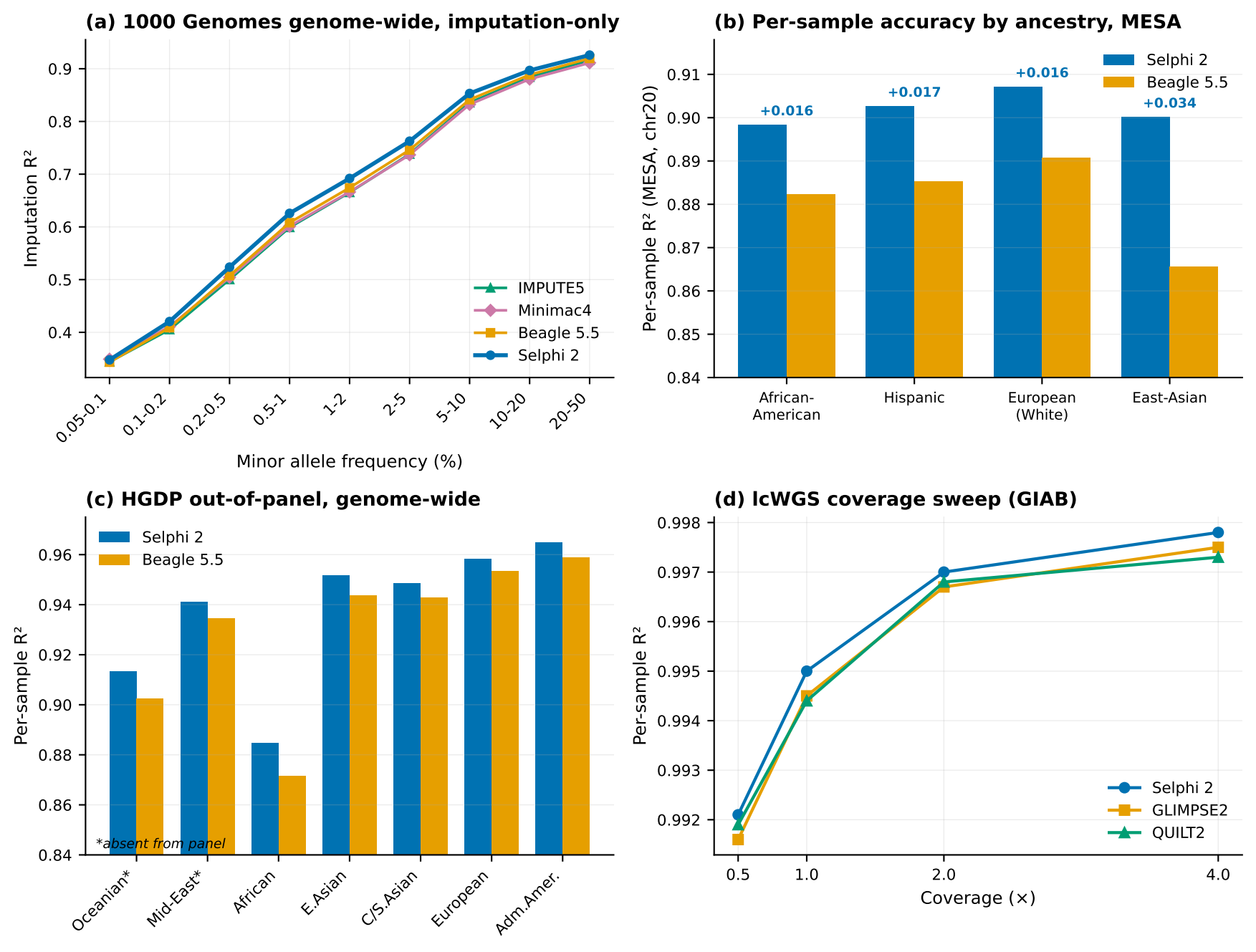
(a) 1000 Genomes genome-wide, imputation-only
Imputation R²
0.9
0.8
0.7
0.6
0.5
0.4
IMPUTE5
Minimac4
Beagle 5.5
Selphi 2
0.05-0.1
0.1-0.2
0.2-0.5
0.5-1
1-2
2-5
5-10
10-20
20-50
Minor allele frequency (%)
(b) Per-sample accuracy by ancestry, MESA
Per-sample R² (MESA, chr20)
0.91
0.90
0.89
0.88
0.87
0.86
0.85
0.84
Selphi 2
Beagle 5.5
+0.016
+0.017
+0.016
+0.034
African-American
Hispanic
European (White)
East-Asian
(c) HGDP out-of-panel, genome-wide
Per-sample R²
0.96
0.94
0.92
0.90
0.88
0.86
0.84
Selphi 2
Beagle 5.5
*absent from panel
Oceanian*
Mid-East*
African
E.Asian
C/S.Asian
European
Adm.Amer.
(d) lcWGS coverage sweep (GIAB)
Per-sample R²
0.998
0.997
0.996
0.995
0.994
0.993
0.992
Selphi 2
GLIMPSE2
QUILT2
0.5
1.0
2.0
4.0
Coverage (×)